SAHAND TABATABAEI

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RESEARCH PROFILE

Quantum control and sensing researcher working across theory, numerical modeling, and experiment. I develop experimentally realistic control and sensing protocols, taking them from analytical modeling and simulation to waveform generation, hardware integration, measurement, and interpretation. My work has focused primarily on spin systems, with broader interests in quantum control, sensing, and Hamiltonian engineering across diverse hardware platforms. I emphasize translating theoretical protocols into practical tools that enable new measurement capabilities in demanding experimental environments.

EDUCATION

PhD, Physics (Quantum Information), University of Waterloo 2020–present

- Thesis: *Quantum Control for Nanoscale Magnetic Resonance: From Quantum Sensing to Atomic-Scale Scattering*
- Supervisor (PhD & MSc): Prof. Raffi Budakian

MSc, Physics, University of Waterloo 2017–2020

- Thesis: *Numerical Engineering of Adiabatic Quantum Operations*

BSc, Electrical Engineering, Iran University of Science and Technology (IUST) 2012–2017

- Thesis: *Optimum Design of Miniaturized Microstrip Travelling-wave-loop Directional Filters*

PUBLICATIONS

Peer-Reviewed

- *Long-lived Mechanically-detected Molecular Spins for Quantum Sensing*, **S. Tabatabaei**, P. Priyadarsi, D. Tay, N. Singh, P. Sahafi, A. Jordan, and R. Budakian, **Phys. Rev. X** 16, 031018 (2026).
- *Large-enhancement Nanoscale Dynamic Nuclear Polarization Near a Silicon Nanowire Surface*, **S. Tabatabaei** (co-first author), P. Priyadarsi, N. Singh, P. Sahafi, D. Tay, A. Jordan, and R. Budakian, **Sci. Adv.** 10, eado9059 (2024).
- *Roadmap on Nanoscale Magnetic Resonance Imaging*, R. Budakian et al., including **S. Tabatabaei**, **Nanotechnology** 35, 412001 (2024).
- *Nuclear Magnetic Resonance Diffraction With Subangstrom Precision*, H. Haas, **S. Tabatabaei** (co-first author), W. Rose, P. Sahafi, M. Piscitelli, A. Jordan, P. Priyadarsi, N. Singh, B. Yager, P. Poole, D. Dalacu, and R. Budakian, **Proc. Natl. Acad. Sci. U.S.A.** 119(40), e2209213119 (2022).
- *Numerical Engineering of Robust Adiabatic Operations*, **S. Tabatabaei**, H. Haas, W. Rose, B. Yager, M. Piscitelli, P. Sahafi, A. Jordan, P. J. Poole, D. Dalacu, and R. Budakian, **Phys. Rev. Applied** 15, 044043 (2021).

RESEARCH EXPERIENCE

Budakian Lab, University of Waterloo 2017–present
Graduate Researcher *Waterloo, ON*

- **Molecular-spin quantum sensing and nanoscale ESR.** Co-developed SQUINT, a quantum-sensing platform combining molecular electron spins, nanomechanical readout, and quantum control. Developed a novel dynamical-decoupling sequence that suppresses electron-electron interactions, extending coherence times by a factor of 60 relative to the Hahn echo time, enabling nanotesla-scale magnetic-field sensing and spectroscopy of small nearby nuclear-spin ensembles. In separate experiments, enhanced nuclear-spin detection sensitivity through electron-to-nuclear polarization transfer, leading to a 200-fold reduction in averaging time.

  Contributions: Pulse-sequence development, low-temperature measurements, data acquisition and analysis, spin-dynamics simulations in Mathematica/Python/JAX, X-band device design in HFSS and VNA characterization.

- **Atomic-resolution magnetic resonance scattering.** Combined high-fidelity dynamical decoupling with intense magnetic-field gradients to encode angstrom-scale spin modulations in nuclear-spin ensembles. These modulations provided a spatial ruler for picometer-precision displacement measurements. In a separate study, used this approach to probe many-body spin transport in an ensemble of approximately 40 million nuclear spins, from the hydrodynamic regime down to the lattice spacing.

  Contributions: Pulse-sequence development, low-temperature measurements, data acquisition and analysis, electronics characterization, spin-dynamics simulations in Mathematica/Python/JAX, and electromagnetic and thermal modeling in COMSOL.

- **Quantum optimal control.** Developed numerical methods for robust quantum-control pulse engineering, incorporating control-field inhomogeneity, spin-spin interactions, and hardware constraints. Co-developed TensorFlow- and JAX-based software leveraging XLA-compiled array operations and automatic differentiation. The framework is now used routinely across the group, including in the above experiments.

- **AWGMax experiment control library.** Led the development of a lab-wide Python library providing a unified framework for experiment definition, waveform generation, simulation, validation, and logging, with interfaces to LabVIEW-based hardware-control software implemented alongside the library. AWGMax is now standard infrastructure used throughout the group's experiments, including ones listed above.

- **Feedback cooling of nanomechanical resonators.** Implemented a MATLAB-based linear-quadratic-Gaussian controller for dual-mode feedback cooling of silicon nanowire resonators, incorporating measured piezo-actuator transfer functions and system delays. The controller is used routinely in the group's experiments, including those above.

Iran University of Science and Technology (IUST) 2016–2017
Undergraduate Researcher *Tehran, Iran*

- **Microstrip directional-filter design.** Developed a numerical design method for phase-velocity-compensated microstrip directional filters and validated the designs using full-wave simulations in CST Microwave Studio and VNA measurements of prototype circuits.

PRESENTATIONS

Research Talks

- *Long-Lived Mechanically-Detected Molecular Spins for Quantum Sensing* (UWaterloo) 2026
- *Atomic-Resolution Magnetic Resonance Scattering: From Structure to Spin Dynamics* (IQC) 2026
- *Direct Measurement of Dipolar Spin Transport Down to Atomic Length Scales* (Cappellaro Group, MIT) 2025
- *Control and Detection of Nanometer-scale Spin Systems: Realizing Nanoscale MR Diffraction* (IQC) 2020

Posters

- *Experimental Study of Dipolar Spin Transport Down to Atomic Length Scales* (NanoMRI7) 2022
  (**Poster award**)
- *Numerical Engineering of Fast Adiabatic Passages* (Spin Canada) 2019

HONORS AND AWARDS

- Marie Curie Graduate Student Award, University of Waterloo 2018–2023
- International Master's and Doctoral Student Awards, University of Waterloo 2018–2025
- Ranked 2nd out of $\sim$ 5000 students in the Iranian national MSc physics entrance exam 2017
- Outstanding Undergraduate Student Award, IUST 2014–2015

TECHNICAL SKILLS

- **Theoretical and Computational**

Theoretical Methods:	Quantum control, pulse-sequence design, optimal control theory, dynamical decoupling, average Hamiltonian theory, adiabatic control, numerical modeling of quantum systems.
Scientific Computing:	Python, JAX, TensorFlow, Git, Mathematica, MATLAB, numerical optimization, statistical data analysis.
Simulation & CAD:	HFSS, COMSOL, CST Microwave Studio, ADS, SPICE.

- **Experimental**

RF/Microwave Engineering:	RF and microwave circuit design, simulation, and characterization. Experience with oscilloscopes, spectrum analyzers, network analyzers, signal generators.
Instrumentation:	Arbitrary waveform generators, multi-channel waveform generation and clock synchronization, lock-in detection, feedback control (PID, LQG), piezo nanopositioning, and automated data acquisition, LabVIEW.
Cryogenics & Vacuum:	Low-temperature experiments at 4 K, LHe and LN_2 handling, high-vacuum systems.

- **Communication Skills**

Scientific Communication:	LaTeX, vector graphics (Inkscape), and 3D scientific rendering (Blender) for publication-quality figures.
Languages:	English (fluent), Persian (native), German (elementary).

MENTORSHIP AND TEACHING EXPERIENCE

Research Mentorship University of Waterloo

- Mentored junior researchers in quantum control, simulation, and experimental measurements, enabling them to conduct measurements independently for their own projects.
- During the PI's one-year sabbatical, assumed primary responsibility for day-to-day experimental progress, troubleshooting, and mentoring junior researchers, while coordinating regularly with the PI remotely.

Course Manager University of Waterloo

Quantum Physics I (3 semesters).

- Co-developed course content with the instructor, including the course schedule, tutorials, assignments, quizzes, and exam questions, and delivered occasional lectures.

Teaching Assistant University of Waterloo

- Responsibilities have included tutorial delivery, in-person office hours, online student support, laboratory instruction, grading, and course logistics.
- *Courses:* Control of Quantum Technologies Laboratory, Mathematical Physics I, Intermediate Classical Mechanics, Light–Matter Interactions, Electricity and Magnetism I, Computational Physics II, Modern Physics Laboratory, Physical Optics, Advanced Calculus for Nano-Engineers II, and General Physics I.

Physics Stack Exchange Contributor

- ∼171k people reached through 145 detailed answers to technical physics questions.

MEDIA COVERAGE

- *Using a Molecule as a Quantum Sensor* (Waterloo News) 2026
- *Waterloo Researchers Advance Nanoscale Imaging Capabilities* (Waterloo News) 2024
- *Subatomic MRI Could Lead to New Drug Therapies* (Waterloo News) 2022